

CPT101 各Lecture结尾Q&A问题整理

Lecture 1 Q&A 问题

1. What does CPU stand for? ALU?
2. What are the components of a computer system?
3. Mention 4 different types of computers.
4. Define 'Downward Compatibility' .
5. What are the advantages of having a hierarchical approach to computer system?
6. What is Moore' s Law?
7. WIMP stand for? WIMP is available due to the development of?

Lecture 2 Q&A 问题

1. Using a diagram, illustrate the concept of Input-Process-Output Model.
2. Which model gives the fundamental structure of the current generation of digital computers?
3. Process is controlled by what?
4. Highlight the three components required for the implementation of Input-Process-Output and von Neumann model.
5. Name 5 examples of computer hardware.
6. Identify the active part within a computer which performs calculations and other operations.
7. Which part of a computer holds data and programs for access by CPU?
8. Give 3 examples of secondary storage.
9. Give 3 examples of input devices.
10. Give 3 examples of output devices.
11. Define 'software' .
12. Difference between Input-Process-Output model & von Neumann model?
13. For a particular machine, the machine instruction set is usually fixed. True or false?
14. There exists a standard instruction set for industry purpose. True or false?
15. High Level Programming Languages (HLLs) are more suitable for programming than the languages of machine instructions. Why?

16. Mention 4 major categories of machine instructions.
17. Why HLL programs need to be translated before execution?
18. What is the von Neumann model?
19. Both ? and ? are held in computer' s memory (store) and both represented by?
20. Identify the von Neumann bottleneck.
21. How could computers distinguish data from instructions since they are both represented by binary codes?
22. What is the main difference between von Neumann machine and Harvard architecture?
23. Motivate the use of Harvard architecture.
24. What is the additional cost from the usage of Harvard architecture?
25. Most desktop CPUs have an internal "instruction cache" feeding the control unit and a completely separate "data cache". This mimicks which computer architecture?

Lecture 3 Q&A 问题

1. Name 4 examples of HLLs.
2. Translation fills in the semantic gap in computer systems. (True or false?)
3. Name 3 different ways of translation. Identify the crucial role of translation under each.
4. When compile-time errors occur, what do we do?
5. What are the purposes of linking?
6. 'Loading' is carried out before 'linking' after a program is compiled. (True or false?)
7. Program modules can be compiled separately. (True or false?)
8. A compiler can translate a module into binary codes, but it cannot resolve those references to other modules. This occurs when ...?
9. Library files are usable if linked into your program code. (True or false?)
10. Interpreters typically convert program code into what?
11. What is the output of program compilation?
12. Name 2 scenarios where interpreters are more useful than compilers.
13. Interpreters are sometimes called as virtual machine because ...?
14. Mention 3 approaches to code sharing.
15. Name 2 disadvantages (or issues) of macro libraries.
16. Libraries can be linked into your program code, but not altered. (True or false?)
17. What are the disadvantages of program execution with pre-translated program library?

18. There exists a de-facto standard of dynamic libraries and dynamic linking. (True or false?)

Lecture 4 Q&A 问题

1. What is an operating system (OS)? What are its core functions?
2. What is the difference between a process and a program?
3. What is process scheduling? What are the common scheduling algorithms?
4. What is memory paging? Why is it used?
5. What is a file system? Give 3 examples of file systems.

Lecture 5 Q&A 问题

1. What is a computer network? What are the key components of a network?
2. What is the difference between LAN, MAN, and WAN?
3. What is TCP/IP protocol suite? What are the functions of TCP and IP?
4. What is an IP address? What is the difference between IPv4 and IPv6?
5. What is a router? What is its role in a network?

Lecture 6 Q&A 问题

1. What is data representation? What are the common ways to represent data in computers?
2. Convert the decimal number 125 to binary and hexadecimal.
3. What is ASCII code? What is the range of ASCII values for uppercase letters, lowercase letters, and digits?

Lecture 7 Q&A 问题

1. What is binary arithmetic? Explain the rules of binary addition and subtraction.
2. How to perform binary multiplication and division?
3. What is two's complement? Why is it used in computers for representing negative numbers?
4. Convert the decimal number -25 to 8-bit two's complement.
5. Perform 8-bit two's complement subtraction: $15 - 7 = ?$

Lecture 8 Q&A 问题

1. What is a logic gate? Name the basic logic gates and their functions.
2. Explain the difference between combinational logic circuits and sequential logic circuits.
3. What is a half-adder? Draw its logic diagram and truth table.

4. What is a full-adder? How is it different from a half-adder?
5. What is a flip-flop? Name the common types of flip-flops.

Lecture 9 Q&A 问题

1. What is computer organization? How is it different from computer architecture?
2. Explain the structure and function of the CPU.
3. What are registers? Name the common CPU registers and their purposes.
4. What is the fetch-decode-execute cycle? Explain each step.
5. How does the control unit coordinate the activities of the CPU?

Lecture 10 Q&A 问题

1. What is memory hierarchy? Explain the different levels of memory hierarchy.
2. What is cache memory? Why is it used in computer systems?
3. Explain the principle of locality of reference and its importance for cache memory.
4. What is the difference between SRAM and DRAM?
5. What is virtual memory? How does it work?

Lecture 11 Q&A 问题

1. What is I/O interface? What are its functions?
2. Explain the different modes of I/O transfer: programmed I/O, interrupt-driven I/O, and DMA.
3. What is an interrupt? How does the CPU handle interrupts?
4. What is DMA (Direct Memory Access)? What is its advantage over interrupt-driven I/O?
5. Name some common I/O devices and their interfaces.

Lecture 12 Q&A 问题

1. What is a programming language? Classify programming languages into different types.
2. Explain the characteristics of procedural programming languages.
3. What is object-oriented programming (OOP)? What are its key concepts?
4. Differentiate between procedural programming and object-oriented programming.
5. Name 3 object-oriented programming languages and their applications.

Lecture 13 Q&A 问题

1. What is a compiler? Explain the phases of compilation.

2. What is an interpreter? How is it different from a compiler?
3. What is an assembler? What is the role of assembly language in programming?
4. Explain the concept of syntax analysis and semantic analysis in compilation.
5. What are the advantages of using a compiler over an interpreter?

Lecture 14 Q&A 问题

1. What is an algorithm? What are the characteristics of a good algorithm?
2. Explain the steps involved in algorithm design.
3. What is pseudocode? How is it used in algorithm representation?
4. Draw a flowchart for finding the factorial of a given number.
5. Differentiate between linear search and binary search algorithms.

Lecture 15 Q&A 问题

1. What is data structure? Why are data structures important in programming?
2. Explain the difference between arrays and linked lists.
3. What is a stack? Explain its operations (push and pop) with examples.
4. What is a queue? Explain its operations (enqueue and dequeue) with examples.
5. Name some applications of stacks and queues.

Lecture 16 Q&A 问题

1. What is a tree data structure? Explain the basic terminology of trees.
2. What is a binary tree? How is it different from a general tree?
3. Explain the in-order, pre-order, and post-order traversal of a binary tree.
4. What is a binary search tree (BST)? What are its properties?
5. How to insert and delete a node in a binary search tree?

Lecture 17 Q&A 问题

1. What is a graph data structure? Explain the types of graphs.
2. How to represent a graph using adjacency matrix and adjacency list?
3. Explain the breadth-first search (BFS) algorithm for graph traversal.
4. Explain the depth-first search (DFS) algorithm for graph traversal.
5. Name some applications of graph data structures.

Lecture 18 Q&A 问题

1. What is sorting? Explain the bubble sort algorithm with an example.
2. Explain the selection sort algorithm and its time complexity.
3. What is insertion sort? How is it different from bubble sort?
4. Explain the merge sort algorithm and its advantages.
5. What is quicksort? Explain its working principle.

Lecture 19 Q&A 问题

1. What is database management system (DBMS)? What are its functions?
2. Explain the difference between file system and DBMS.
3. What is relational database? What are the characteristics of relational model?
4. What is SQL? What are the different types of SQL commands?
5. Write an SQL query to retrieve data from a table with a WHERE clause.

Lecture 20 Q&A 问题

1. What is normalization? Why is it important in database design?
2. Explain the first normal form (1NF) and second normal form (2NF).
3. What is third normal form (3NF)? How to achieve it?
4. Explain the concept of primary key, foreign key, and unique key.
5. What is database indexing? What are its advantages and disadvantages?

Lecture 21 Q&A 问题

1. What is computer security? What are the main threats to computer security?
2. Explain the concept of authentication and authorization.
3. What is encryption? Differentiate between symmetric and asymmetric encryption.
4. What is a firewall? How does it protect a computer network?
5. Name some common security measures to protect computer systems.

Lecture 22 Q&A 问题

1. What is cloud computing? What are the service models of cloud computing?
2. Explain the difference between public cloud, private cloud, and hybrid cloud.
3. What are the advantages and disadvantages of cloud computing?

4. Name some popular cloud computing service providers and their services.
5. What is virtualization in cloud computing? How does it work?

Lecture 23 Q&A 问题

1. What is artificial intelligence (AI)? How is it different from human intelligence?
2. Explain the types of artificial intelligence: narrow AI and general AI.
3. What is machine learning? What are the main types of machine learning?
4. Explain the concept of neural networks and their applications.
5. Name some real-world applications of artificial intelligence.

Lecture 24 Q&A 问题

1. What is the future of computer technology? Discuss some emerging trends.
2. Explain the impact of computer technology on society and economy.
3. What are the ethical issues related to the development and use of computer technology?
4. Discuss the importance of lifelong learning in the field of computer science.
5. What are the key skills required for a career in computer science?

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